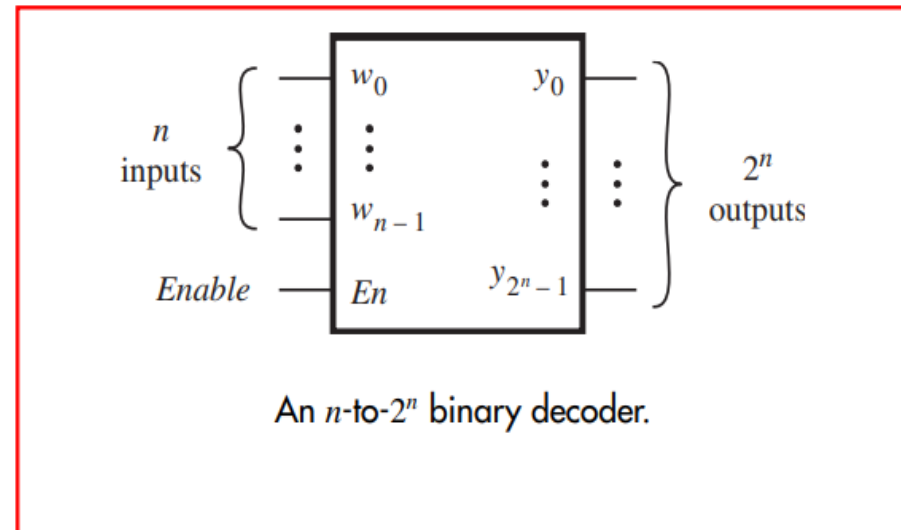


Digital Electronics
III Semester
B. Tech. (CSE)
Amrita School of Engineering, Bengaluru

Decoder

- Decoder circuits are used to decode encoded information.
- A binary decoder is a logic circuit with n inputs and 2^n outputs.
- Only one output is asserted at a time, and each output corresponds to one valuation of the inputs.
- The decoder also has an enable input, En , that is used to disable the outputs;
 - If $En = 0$, then none of the decoder outputs is asserted.
 - If $En = 1$, the valuation of $w_{n-1} \cdots w_1 w_0$ determines which of the outputs is asserted.



Decoder

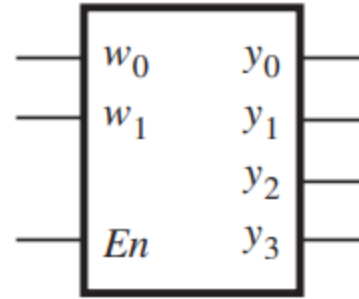
- An n-bit binary code in which exactly one of the bits is set to 1 at a time is referred to as one-hot encoded, meaning that the single bit that is set to 1 is deemed to be “hot.”
- The outputs of a binary decoder are one-hot encoded.
- A decoder can be designed to have either active-high or active-low outputs.

Decoder

2-to-4 Decoder (Active High Output)-

En	w_1	w_0	y_0	y_1	y_2	y_3
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1
0	x	x	0	0	0	0

(a) Truth table



(b) Graphical symbol

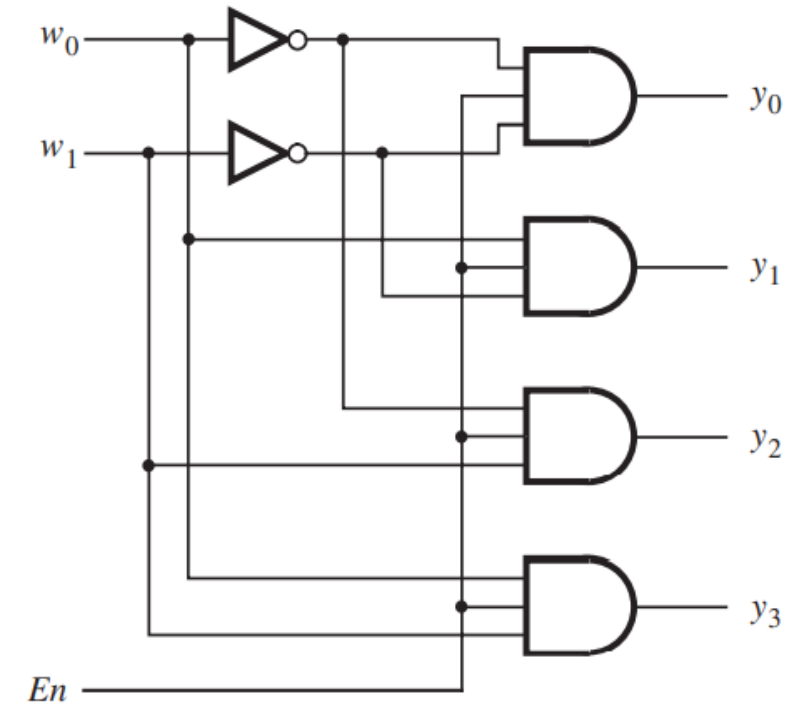
Logical Expressions-

$$y_0 = En \cdot w_1' \cdot w_0'$$

$$y_1 = En \cdot w_1' \cdot w_0$$

$$y_2 = En \cdot w_1 \cdot w_0'$$

$$y_3 = En \cdot w_1 \cdot w_0$$



(c) Logic circuit

Decoder

2-to-4 Decoder (Active Low Output)-

E	A	B	D_0	D_1	D_2	D_3
1	X	X	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	0

(a) Truth table

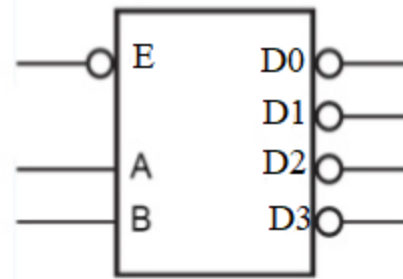
Logical Expressions-

$$D_0 = \overline{E' \cdot A' \cdot B'}$$

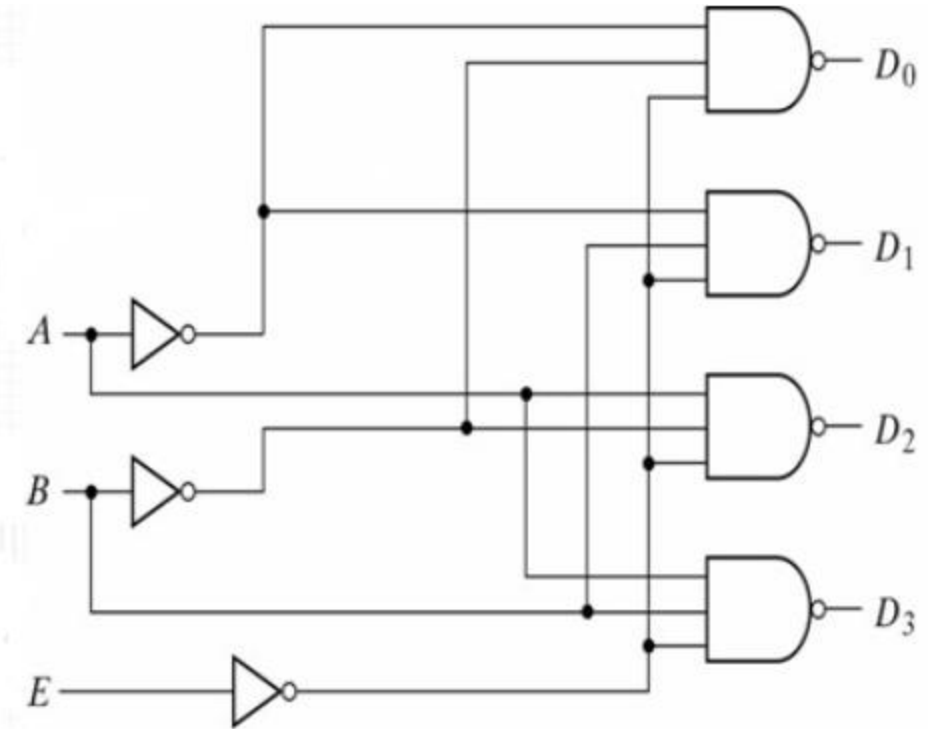
$$D_1 = \overline{E' \cdot A' \cdot B}$$

$$D_2 = \overline{E' \cdot A \cdot B'}$$

$$D_3 = \overline{E' \cdot A \cdot B}$$



(b) Graphical symbol



(c) Logic circuit

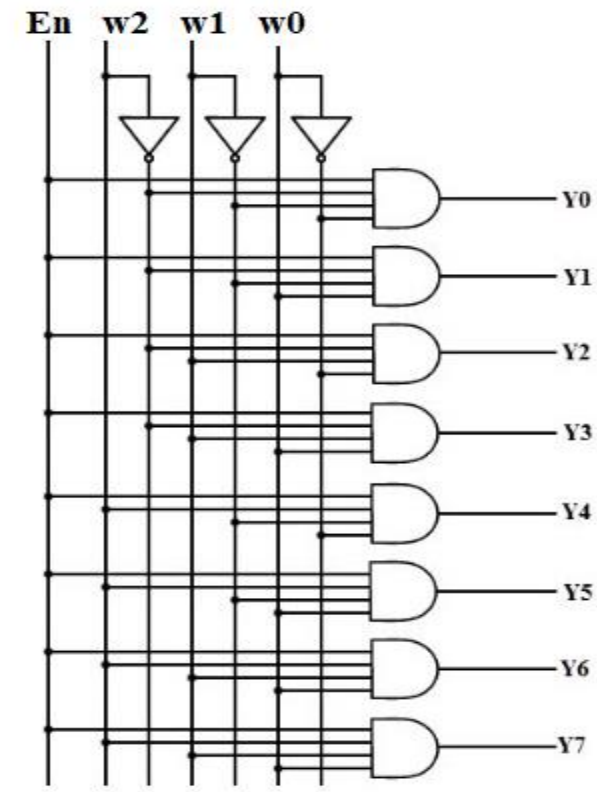
Decoder

3-to-8 Decoder-

- Larger decoders can be built using the sum-of-products structure as shown for 2-to-4 decoder.

Inputs				Outputs							
EN	w ₂	w ₁	w ₀	y ₇	y ₆	y ₅	y ₄	y ₃	y ₂	y ₁	y ₀
0	x	x	x	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	1
1	0	0	1	0	0	0	0	0	0	1	0
1	0	1	0	0	0	0	0	0	1	0	0
1	0	1	1	0	0	0	0	1	0	0	0
1	1	0	0	0	0	0	1	0	0	0	0
1	1	0	1	0	0	1	0	0	0	0	0
1	1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0	0	0

Truth Table

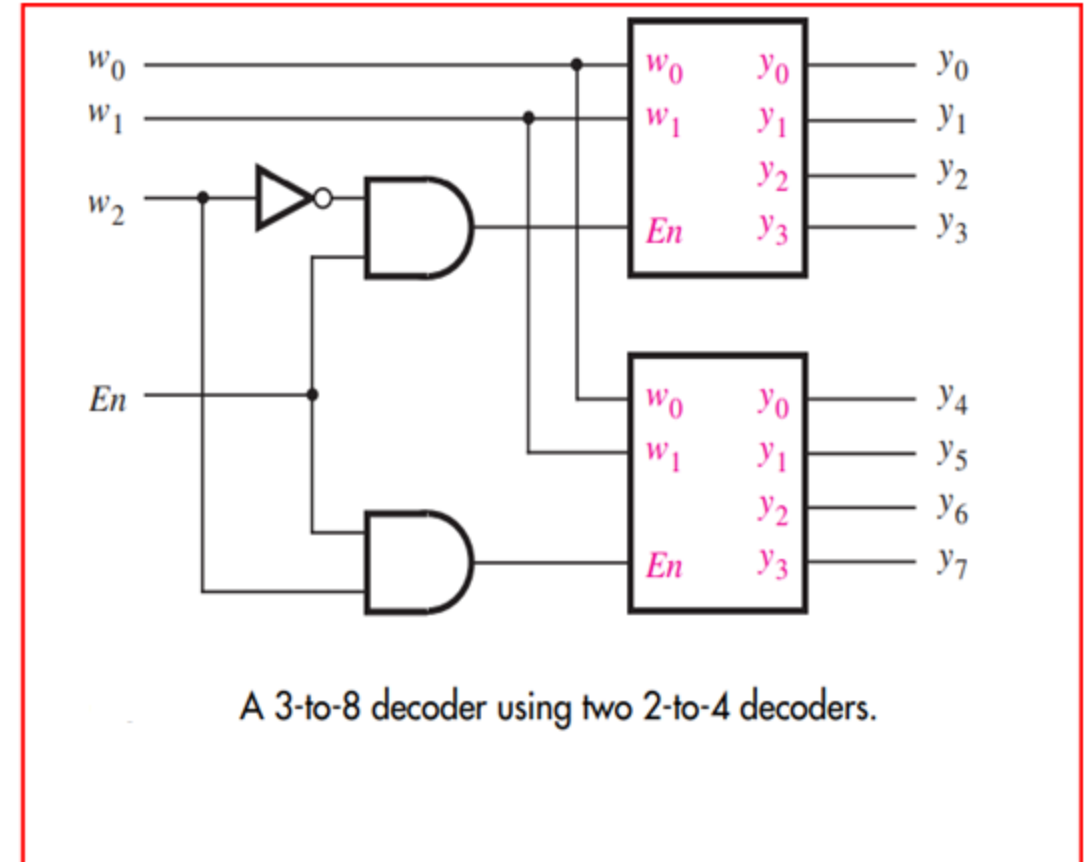


Logic Circuit

Decoder

3-to-8 Decoder-

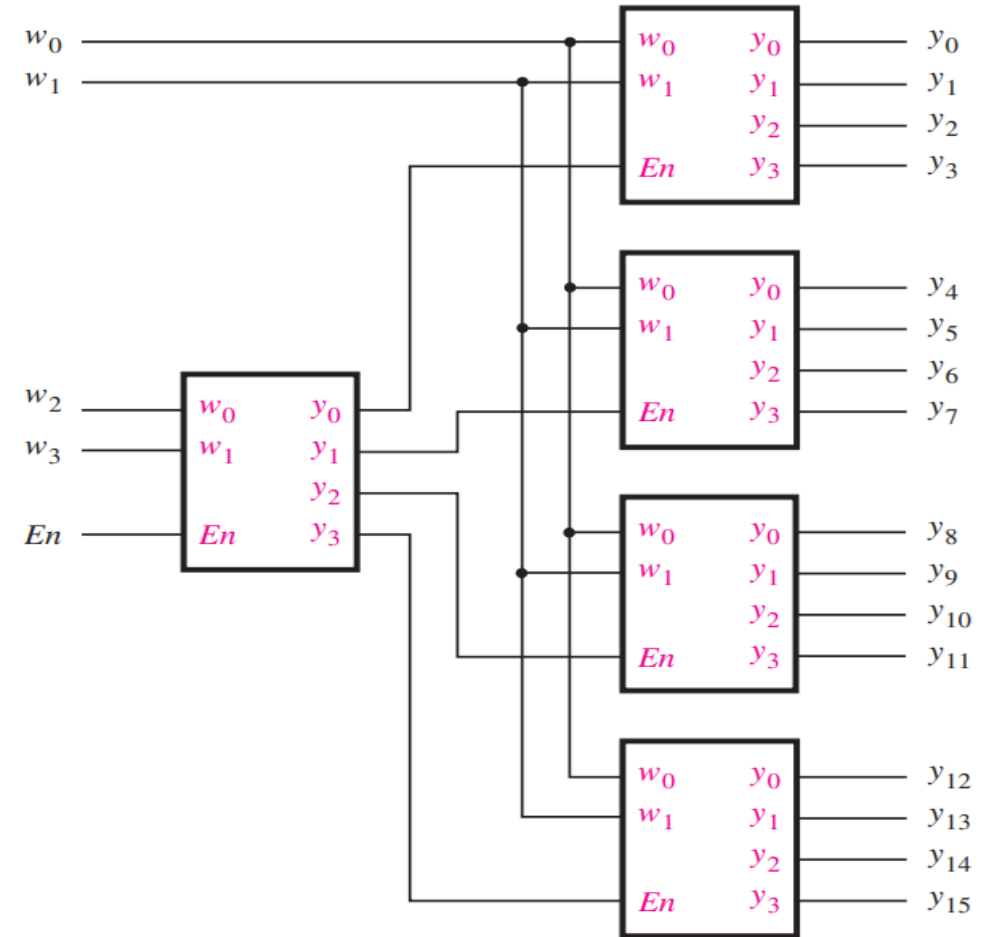
- A larger decoder can be constructed from smaller decoders.
- Figure shows how a 3-to-8 decoder is built with two 2-to-4 decoders.
- The w_2 input drives the enable inputs of the two decoders. The top decoder is enabled if $w_2 = 0$, and the bottom decoder is enabled if $w_2 = 1$.
- This concept can be applied for decoders of any size.



Decoder

4-to-16 Decoder-

- Five 2-to-4 decoders can be used to construct a 4-to-16 decoder.
- Because of its treelike structure, this type of circuit is often referred to as a decoder tree.

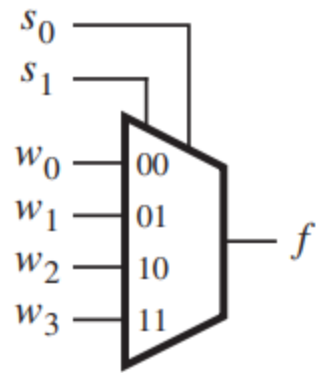


A 4-to-16 decoder built using a decoder tree.

Decoder

4-to-1 Multiplexer using Decoder-

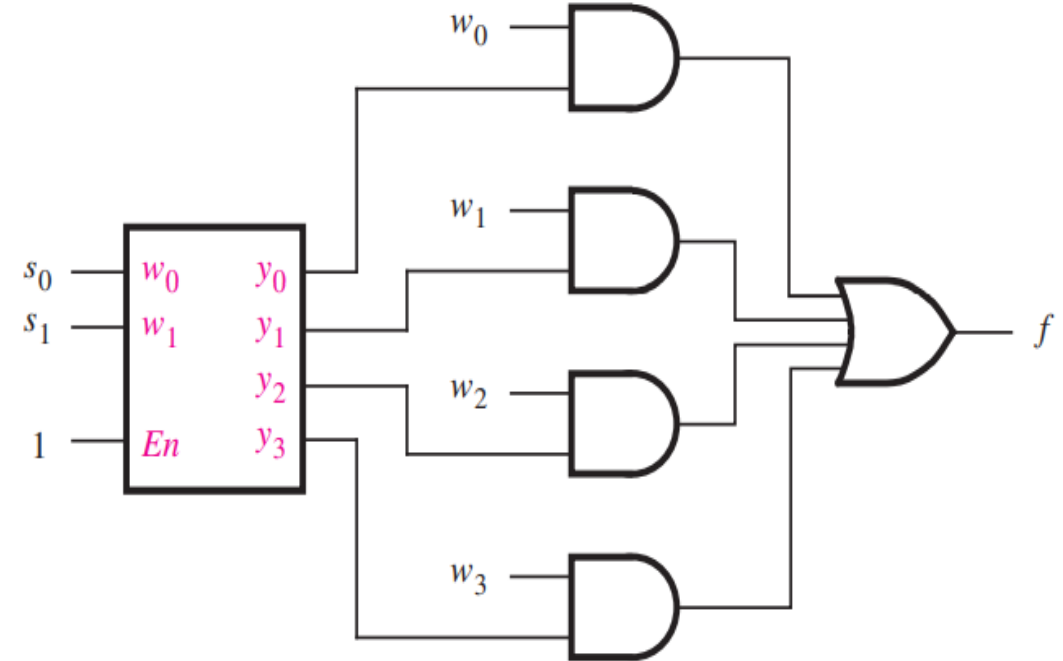
4-to-1 Multiplexer



(a) Graphical symbol

s_1	s_0	f
0	0	w_0
0	1	w_1
1	0	w_2
1	1	w_3

(b) Truth table



A 4-to-1 multiplexer built using a decoder.

Decoder

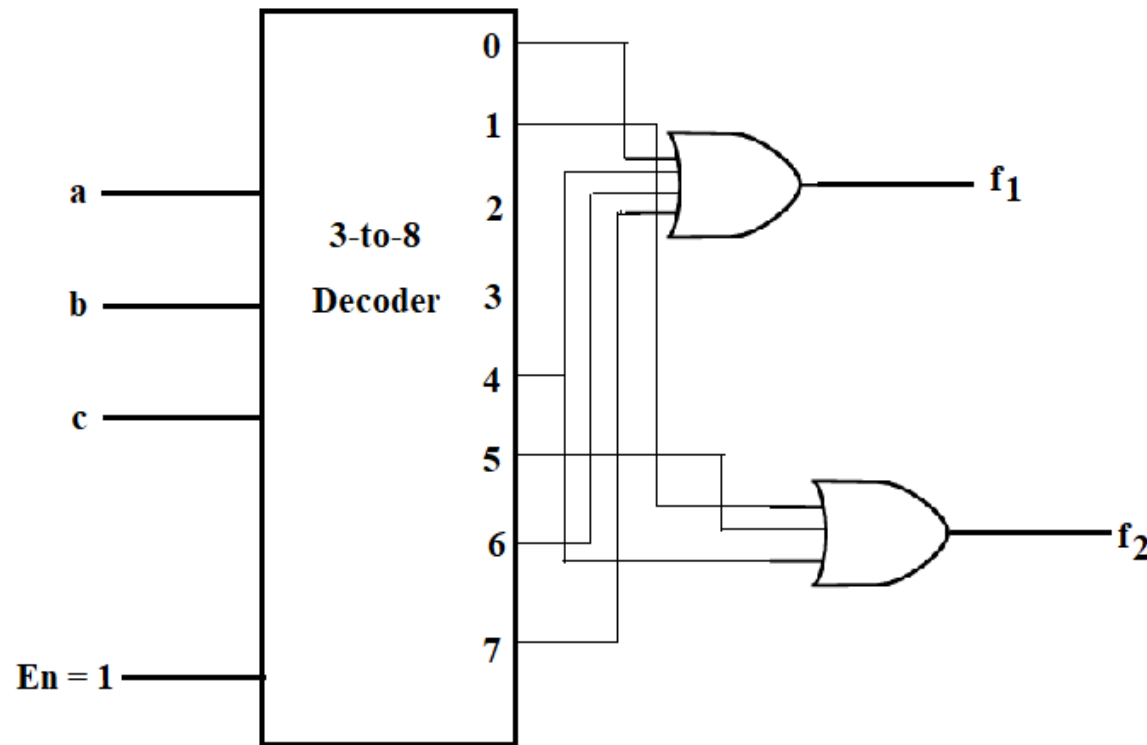
Function Implementation using Decoder (Active High Output)-

SOP Form-

Implement the following functions using a 3 to 8 line decoder and required logic gates.

$$f_1(a, b, c) = \sum m(0, 4, 6, 7)$$

$$f_2(a, b, c) = \sum m(1, 4, 5)$$



Decoder

Function Implementation using Decoder (Active High Output)-

POS Form-

Implement the following functions using a decoder minimizing the number of inputs to be summed.

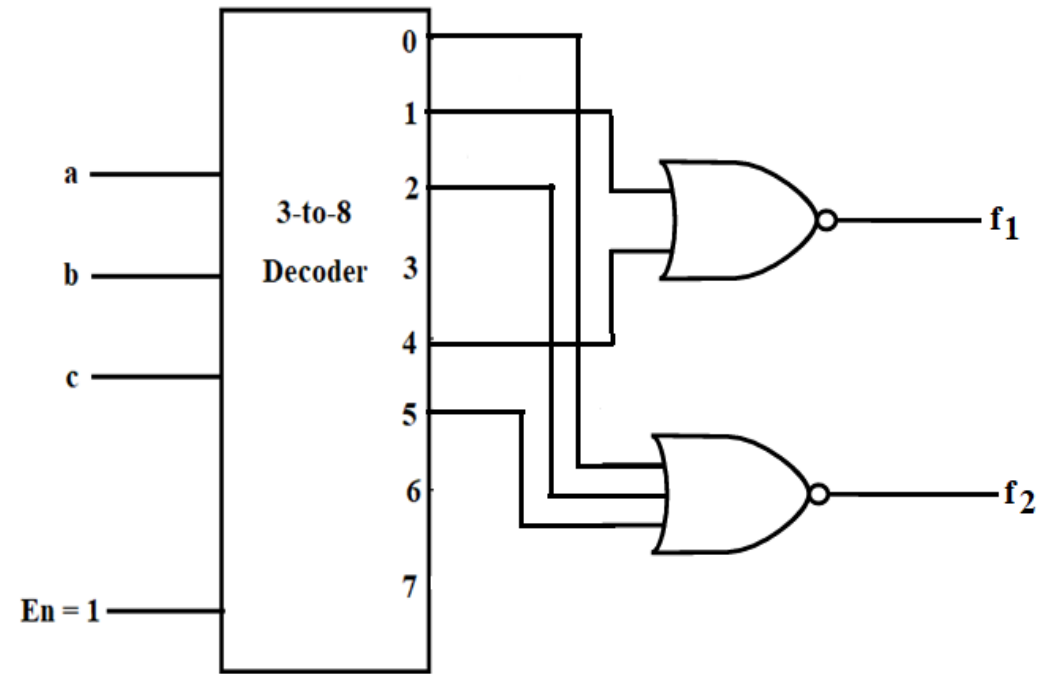
$$f_1(a, b, c) = \sum m(0, 2, 3, 5, 6, 7)$$

$$f_2(a, b, c) = \sum m(1, 3, 4, 6, 7)$$

Solution- In POS form-

$$f_1(a, b, c) = \prod M(1, 4) = \overline{\sum m(1, 4)}$$

$$f_2(a, b, c) = \prod M(0, 2, 5) = \overline{\sum m(0, 2, 5)}$$



Decoder

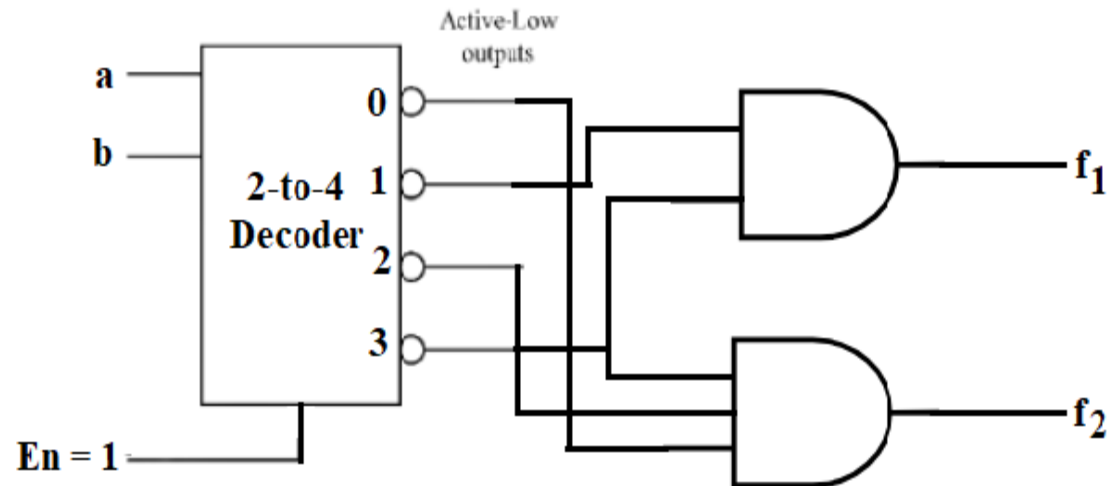
Function Implementation using Decoder (Active Low Output)-

POS Form-

Implement the following functions in maxterm canonical form using a decoder with active low outputs and required logic gates.

$$f_1(a, b) = \Pi M (1,3)$$

$$f_2(a, b) = \Pi M (0,2,3)$$



Decoder

Function Implementation using Decoder (Active Low Output)-

SOP Form-

Implement the following functions using a 3 to 8 decoder minimizing the number of inputs to the logic gates with active low outputs.

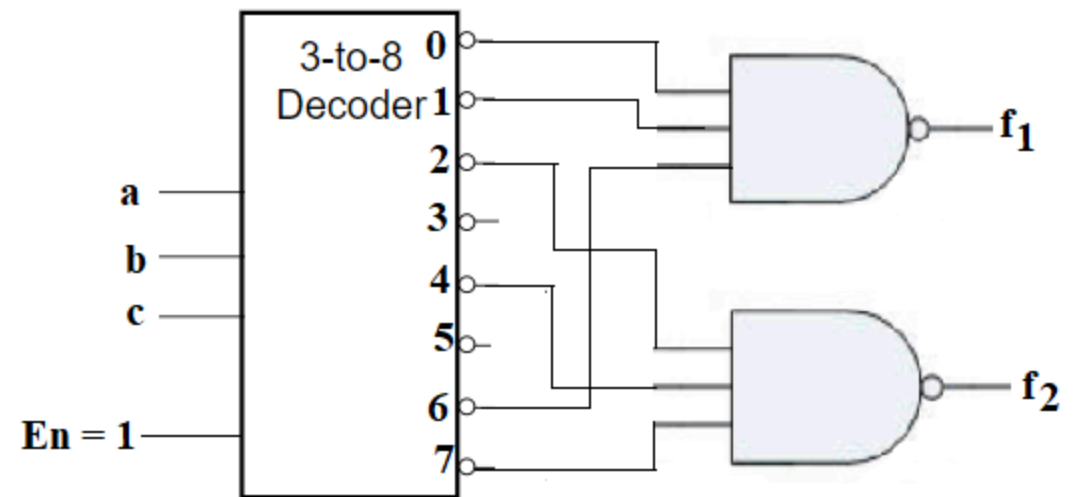
$$f_1(a, b, c) = \Pi M (0,1,3,5,6)$$

$$f_2(a, b, c) = \Pi M (2,3,4,5,7)$$

Solution- In SOP form

$$f_1(a, b, c) = \Sigma m (2,4,7) = \overline{\Pi M (2,4,7)}$$

$$f_2(a, b, c) = \Sigma m (0,1,6) = \overline{\Pi M (0,1,6)}$$



Decoder

Function Implementation using Decoder-

Implement full adder using a decoder and required logic gates.

Solution- To implement a full adder, we need a 3:8 decoder and OR gates. The input to the full adder, first and second bits and carry bit, are used as input to the decoder. Let x, y and z represent these three bits. Sum and Carry outputs of a full adder have the following truth table-

x	y	z	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$S = \sum(1, 2, 4, 7)$$
$$C = \sum(3, 5, 6, 7)$$

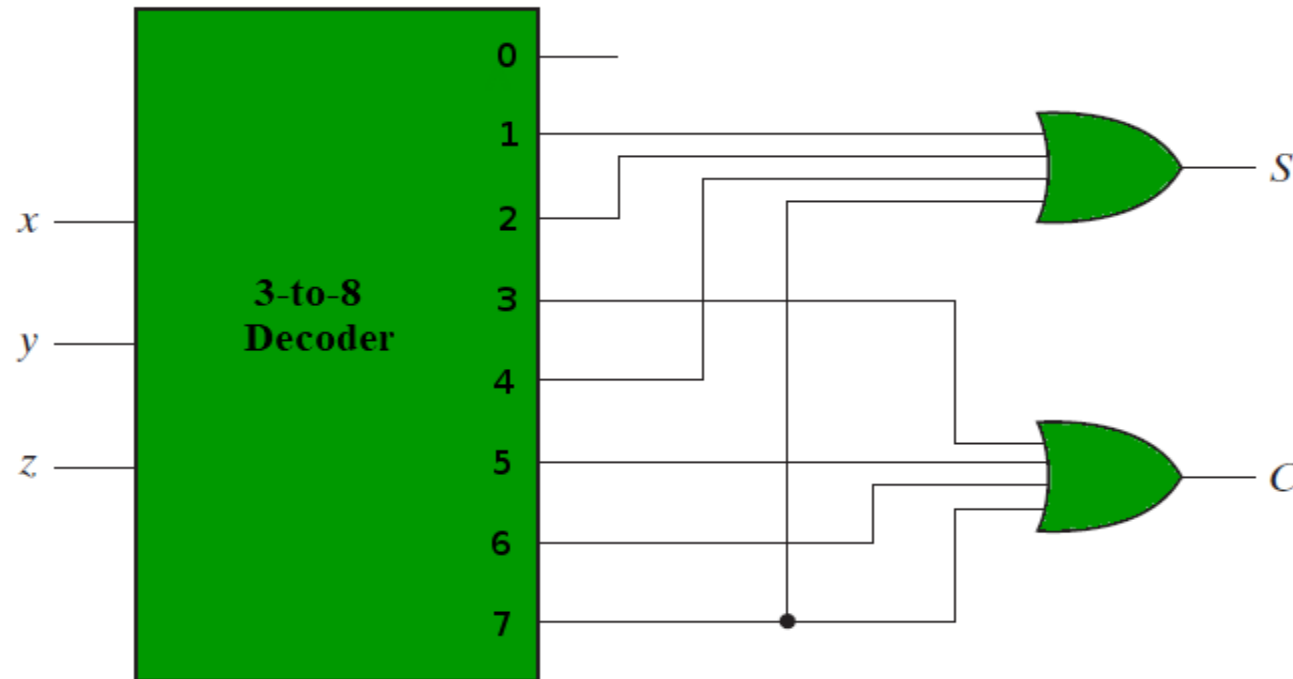
Decoder

Function Implementation using Decoder-

Implement full adder using a decoder and required logic gates.

$$S = \sum(1, 2, 4, 7)$$

$$C = \sum(3, 5, 6, 7)$$



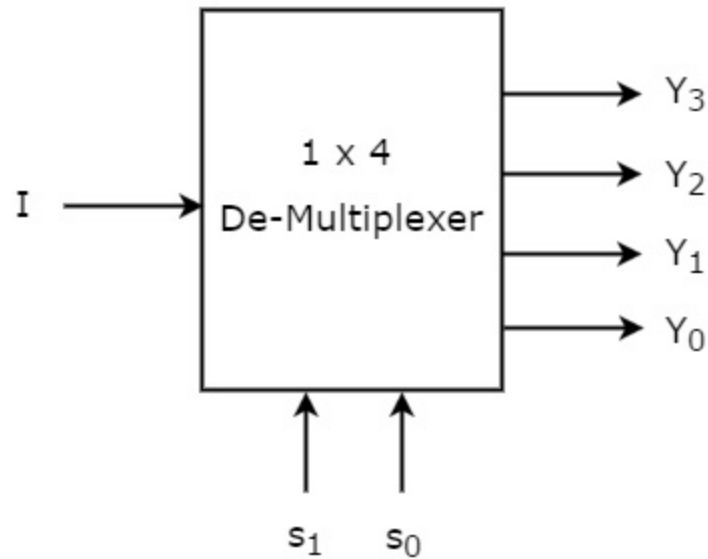
Demultiplexer

- **De-Multiplexer** is a combinational circuit that performs the reverse operation of Multiplexer.
- It has single input, 'n' selection lines and maximum of 2^n outputs. The input will be connected to one of these outputs based on the values of selection lines.
- Since there are 'n' selection lines, there will be 2^n possible combinations of zeros and ones. So, each combination can select only one output.
- The demultiplexer can be implemented using a decoder circuit.
- In general, an n-to- 2^n decoder circuit can be used as a 1-to-n demultiplexer.

Demultiplexer

1-to-4 Demultiplexer

- 1x4 De-Multiplexer has one input I , two selection lines, s_1 & s_0 and four outputs Y_3 , Y_2 , Y_1 & Y_0 .



Block Diagram

Selection Inputs		Outputs			
s_1	s_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0

Truth Table

$$Y_3 = s_1 s_0 I$$

$$Y_2 = s_1 s_0' I$$

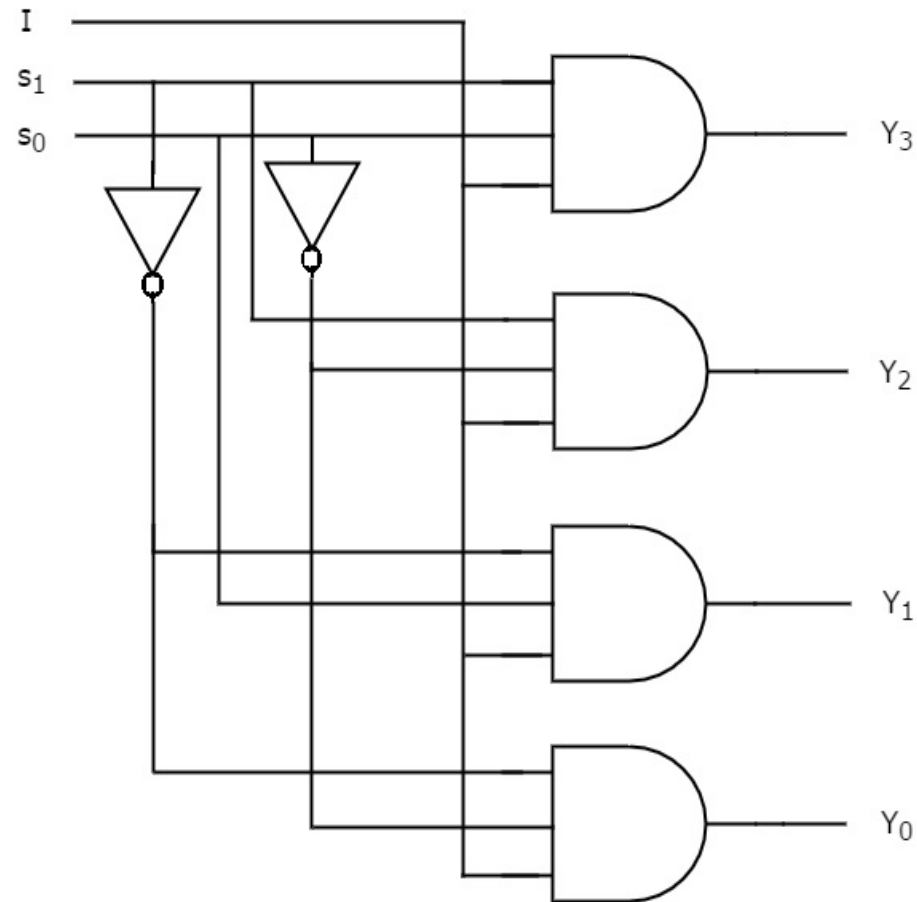
$$Y_1 = s_1' s_0 I$$

$$Y_0 = s_1' s_0' I$$

Logical
Expression

Demultiplexer

1-to-4 Demultiplexer

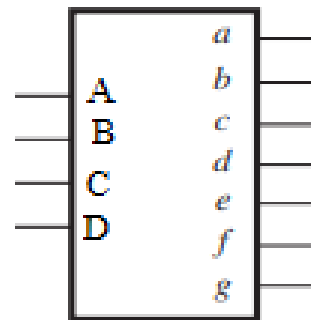


Logic Circuit

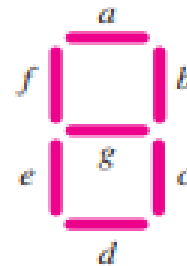
Code converter

BCD to 7-segment Display decoder

- A BCD-to-7-segment decoder converts one binary-coded decimal (BCD) digit into information suitable for driving a digit-oriented display.
- The circuit converts the BCD digit into seven signals that are used to drive the segments in the display.
- Each segment is a small light-emitting diode (LED), which glows when driven by an electrical signal. The segments are labeled from a to g.



(a) Code converter



(b) 7-segment display

Code converter

BCD to 7-segment Display decoder

Truth Table

Decimal Digit	Input lines				Output lines							Display pattern
	A	B	C	D	a	b	c	d	e	f	g	
0	0	0	0	0	1	1	1	1	1	1	0	0
1	0	0	0	1	0	1	1	0	0	0	0	1
2	0	0	1	0	1	1	0	1	1	0	1	2
3	0	0	1	1	1	1	1	1	0	0	1	3
4	0	1	0	0	0	1	1	0	0	1	1	4
5	0	1	0	1	1	0	1	1	0	1	1	5
6	0	1	1	0	1	0	1	1	1	1	1	6
7	0	1	1	1	1	1	1	0	0	0	0	7
8	1	0	0	0	1	1	1	1	1	1	1	8
9	1	0	0	1	1	1	1	1	0	1	1	9

$$a = \sum m(0,2,3,5,6,7,8,9) + d(10,11,12,13,14,15)$$

$$b = \sum m(0,1,2,3,4,7,8,9) + d(10,11,12,13,14,15)$$

$$c = \sum m(0,1,3,4,5,6,7,8,9) + d(10,11,12,13,14,15)$$

$$d = \sum m(0,2,3,5,6,8,9) + d(10,11,12,13,14,15)$$

$$e = \sum m(0,2,6,8) + d(10,11,12,13,14,15)$$

$$f = \sum m(0,4,5,6,8,9) + d(10,11,12,13,14,15)$$

$$g = \sum m(2,3,4,5,6,8,9) + d(10,11,12,13,14,15)$$

Code converter

BCD to 7-segment Display decoder

K-map Simplification-

K-map for a

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	x	x	x	x
10	1	1	x	x

$$a = A + C + BD + \bar{B}\bar{D}$$

K-map for b

AB \ CD	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	x	x	x	x
10	1	1	x	x

$$b = \bar{B} + \bar{C}\bar{D} + CD$$

K-map for c

AB \ CD	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	x	x	x	x
10	1	1	x	x

$$c = B + \bar{C} + D$$

K-map for d

AB \ CD	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	x	x	x	x
10	1	1	x	x

$$d = \bar{B}\bar{D} + C\bar{D} + B\bar{C}D + \bar{B}C + A$$

K-map for e

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	x	x	x	x
10	1	0	x	x

$$e = \bar{B}\bar{D} + C\bar{D}$$

K-map for f

AB \ CD	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

$$f = A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$$

K-map for g

AB \ CD	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	x	x	x	x
10	1	1	x	x

$$g = A + B\bar{C} + \bar{B}C + C\bar{D}$$

Code converter

BCD to 7-segment Display decoder

Logic Circuit

